

ICT2217 Network Security

Lab Exercise 3.3: AAA Part 2 - Authorization

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Configure authorization using privilege levels
- Configure authorization using role-based CLI views

Required Hardware

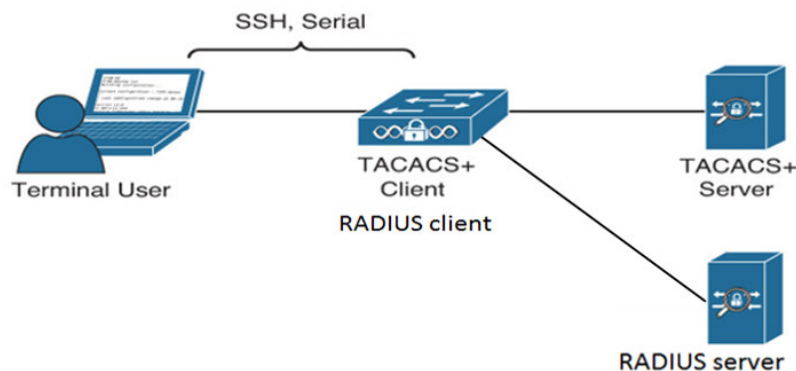
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive
- TACACS.net 2.1.2 Demo version for Windows
<https://www.softpedia.com/get/Internet/Servers/Other-Servers/TACACS-net.shtml>
- FreeRADIUS 3.2.3 for Kali Linux

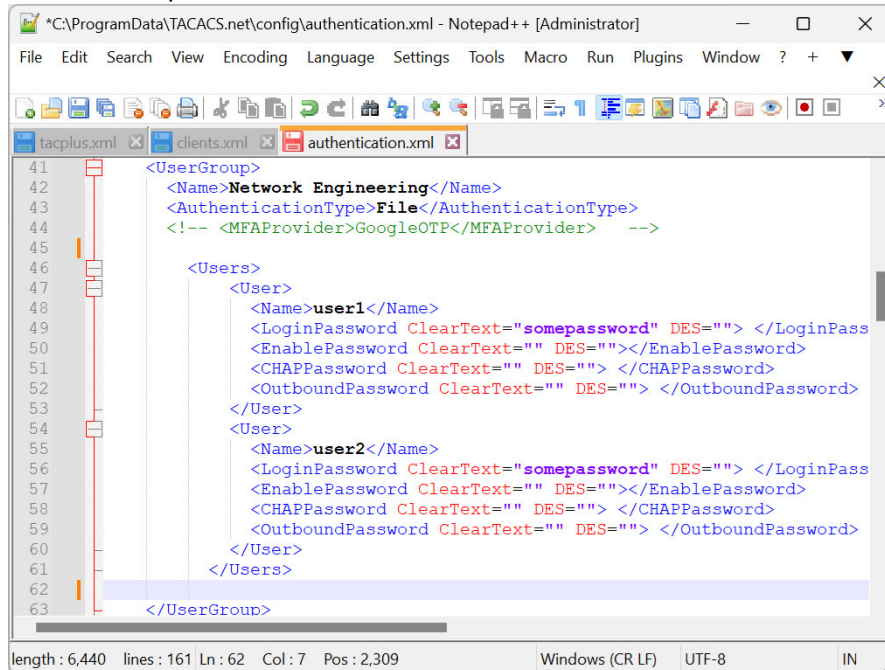
PART 1: Configuring Authorization using Privilege Levels

- 1.1 Note that AAA Authorization will only work when AAA Authentication is also configured. Hence, set up and re-configure the same network topology as done in Lab Exercise 3.2 Part 3.



- 1.2 Follow the example in Lab 3.3 Notes pg 5 to assign some commands of your choice to different privilege level.

- 1.3 To assign privilege level to users locally on the network device, follow the examples in Lab 3.3 Notes pg 7.
- 1.4 To assign privilege level to users in centralised TACACS+ server, remember to first configure user authentication in the **authentication.xml** file as done in Lab Exercise 3.2 Part 2 re-produced here:

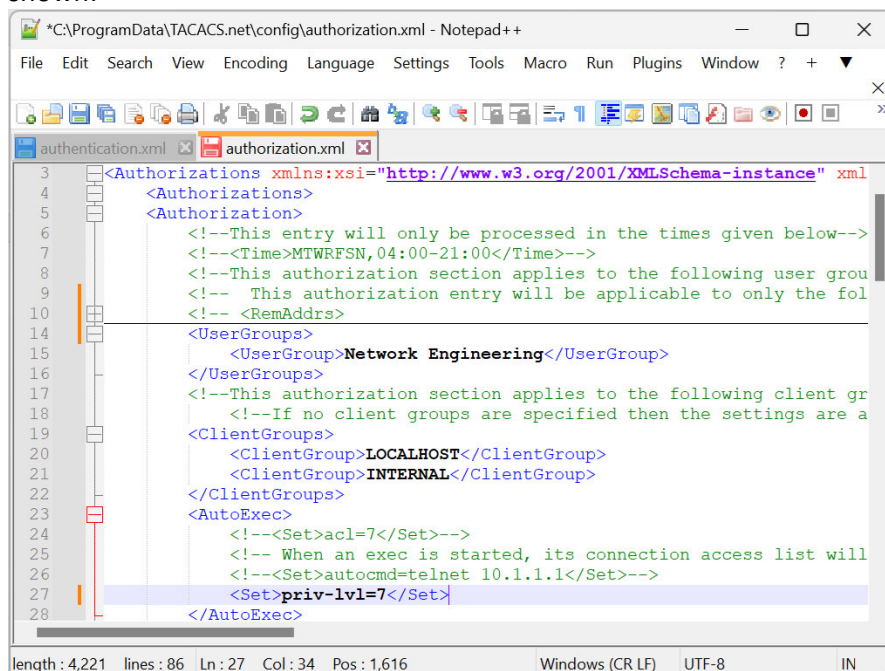


```

41 <UserGroup>
42   <Name>Network Engineering</Name>
43   <AuthenticationType>File</AuthenticationType>
44   <!-- <MFAPProvider>GoogleOTP</MFAPProvider> -->
45
46   <Users>
47     <User>
48       <Name>user1</Name>
49       <LoginPassword ClearText="somepassword" DES=""></LoginPass
50       <EnablePassword ClearText="" DES=""></EnablePassword>
51       <CHAPPassword ClearText="" DES=""></CHAPPassword>
52       <OutboundPassword ClearText="" DES=""></OutboundPassword>
53     </User>
54     <User>
55       <Name>user2</Name>
56       <LoginPassword ClearText="somepassword" DES=""></LoginPass
57       <EnablePassword ClearText="" DES=""></EnablePassword>
58       <CHAPPassword ClearText="" DES=""></CHAPPassword>
59       <OutboundPassword ClearText="" DES=""></OutboundPassword>
60     </User>
61   </Users>
62 </UserGroup>
63
length: 6,440 lines: 161 Ln: 62 Col: 7 Pos: 2,309 Windows (CR LF) UTF-8 IN

```

Then open the corresponding **authorization.xml** file to assign the privilege level as shown:

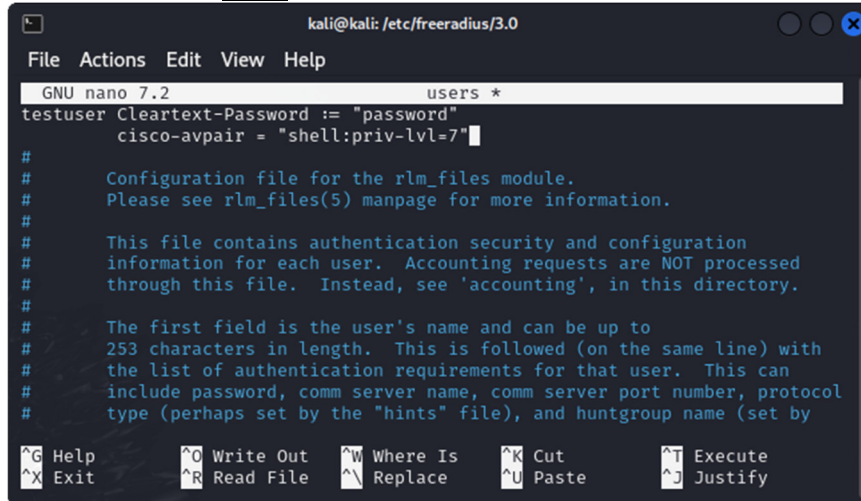


```

3 <Authorizations xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xml
4 <Authorizations>
5 <Authorization>
6   <!--This entry will only be processed in the times given below-->
7   <!--<Time>MTWRFSSN,04:00-21:00</Time>-->
8   <!--This authorization section applies to the following user grou
9   <!-- This authorization entry will be applicable to only the fol
10  <!-- <RemAddrs>
11
12  <UserGroups>
13    <UserGroup>Network Engineering</UserGroup>
14  </UserGroups>
15
16  <!--This authorization section applies to the following client gr
17  <!--If no client groups are specified then the settings are a
18  <ClientGroups>
19    <ClientGroup>LOCALHOST</ClientGroup>
20    <ClientGroup>INTERNAL</ClientGroup>
21  </ClientGroups>
22
23  <AutoExec>
24    <!--<Set>acl=7</Set>-->
25    <!-- When an exec is started, its connection access list will
26    <!--<Set>autocmd=telnet 10.1.1.1</Set>-->
27    <Set>priv-lvl=7</Set>
28  </AutoExec>

```

- 1.5 To assign privilege level to users in centralised RADIUS server, include the privilege level in the same **users** file for authentication as shown:



```

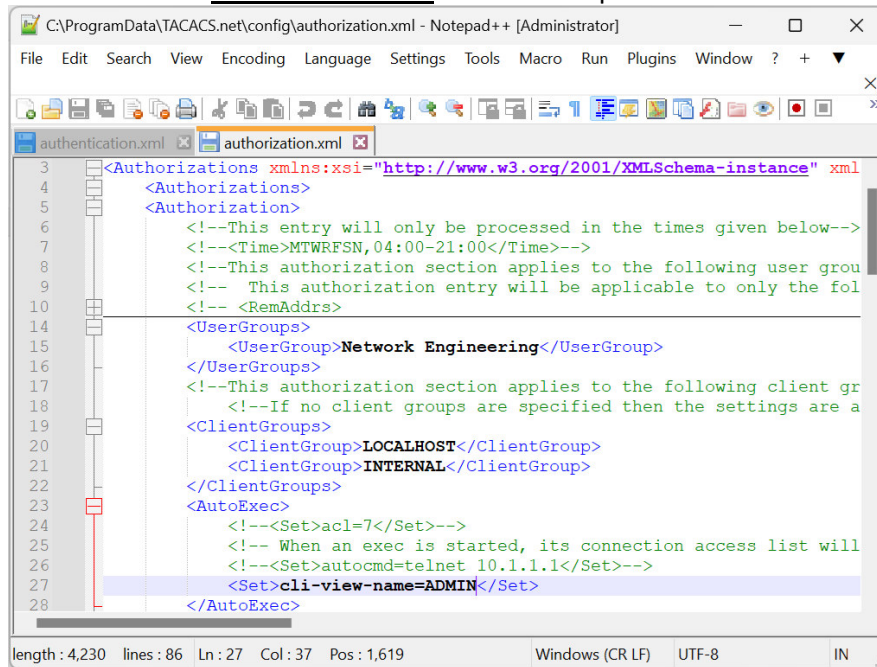
kali@kali: /etc/freeradius/3.0
File Actions Edit View Help
GNU nano 7.2 users *
testuser Cleartext-Password := "password"
        cisco-avpair = "shell:priv-lvl=7"
#
# Configuration file for the rlm_files module.
# Please see rlm_files(5) manpage for more information.
#
# This file contains authentication security and configuration
# information for each user. Accounting requests are NOT processed
# through this file. Instead, see 'accounting', in this directory.
#
# The first field is the user's name and can be up to
# 253 characters in length. This is followed (on the same line) with
# the list of authentication requirements for that user. This can
# include password, comm server name, comm server port number, protocol
# type (perhaps set by the "hints" file), and huntgroup name (set by
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
  
```

- 1.6 Finally, enable AAA authorization as shown in Lab 3.3 Notes pg 8.
- 1.7 For experiment, configure different privilege level for TACACS+ server, RADIUS server and local login.
- 1.8 When done, enter 'exit' until you are disconnected.
- 1.9 Now, login again using console connection or SSH. What is your privilege level after login?
- 1.10 Are you able to execute commands of the same or lower privilege levels?
- 1.11 Are you able to execute commands of higher privilege levels?
- 1.12 Simulate TACACS+ server is down by disconnecting it from the switch, then try login again. What is your privilege level after login?
- 1.13 Next, simulate RADIUS server is down also by disconnecting it from the switch, then try login again. What is your privilege level after login?

PART 2: Configuring Authorization using Role-Based CLI Views

- 2.1 Continue with the same network topology in Part 1. Remember to connect back TACACS+ and RADIUS servers.
- 2.2 Follow the examples in Lab 3.3 Notes pg 11 - 14, create CLI views of your choice and configure some commands into them.

- 2.3 Next, follow the examples in Lab 3.3 Notes pg 15 - 16, create supervIEWS of your choice and configure some CLI views that you've configured in Step 2.2 into them.
- 2.4 Finally, assign views to users as shown in Lab 3.3 Notes pg 18.
- 2.5 To assign views to users in the centralised TACACS+ server, replace privilege level with view in the authorization.xml file. An example as shown:

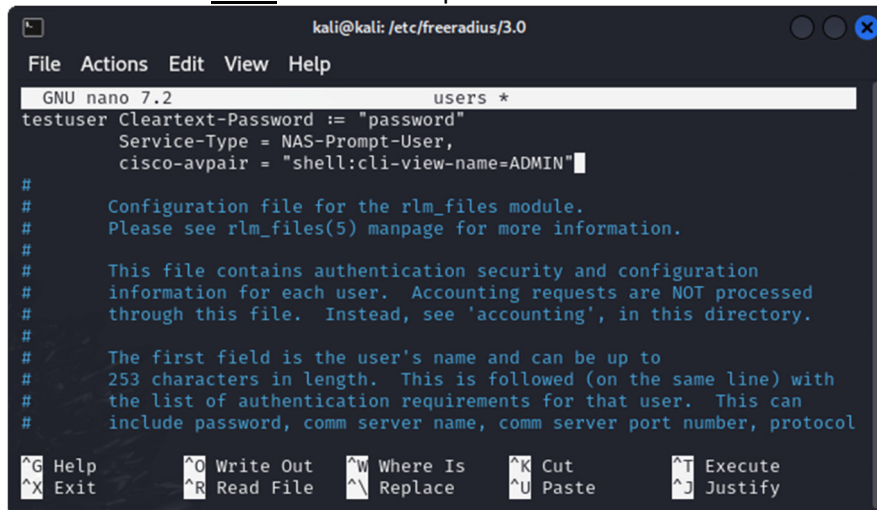


```

3  <Authorizations xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xml
4  <Authorizations>
5  <Authorization>
6      <!--This entry will only be processed in the times given below-->
7      <!--<Time>MTWRFSSN,04:00-21:00</Time>-->
8      <!--This authorization section applies to the following user grou
9      <!-- This authorization entry will be applicable to only the fol
10     <!-- <RemAddr>
11
12
13     <UserGroups>
14         <UserGroup>Network Engineering</UserGroup>
15     </UserGroups>
16
17     <!--This authorization section applies to the following client gr
18     <!--If no client groups are specified then the settings are a
19     <ClientGroups>
20         <ClientGroup>LOCALHOST</ClientGroup>
21         <ClientGroup>INTERNAL</ClientGroup>
22     </ClientGroups>
23     <AutoExec>
24         <!--<Set>acl=7</Set>-->
25         <!-- When an exec is started, its connection access list will
26         <!--<Set>autocmd=telnet 10.1.1.1</Set>-->
27         <Set>cli-view-name=ADMIN</Set>
28     </AutoExec>

```

- 2.6 Similarly, to assign views to users in the centralised RADIUS server, replace privilege level with view in the users file. An example as shown:



```

kali@kali: /etc/freeradius/3.0
File Actions Edit View Help
GNU nano 7.2 users *
testuser Cleartext-Password := "password"
Service-Type = NAS-Prompt-User,
cisco-avpair = "shell:cli-view-name=ADMIN"

#
# Configuration file for the rlm_files module.
# Please see rlm_files(5) manpage for more information.
#
# This file contains authentication security and configuration
# information for each user. Accounting requests are NOT processed
# through this file. Instead, see 'accounting', in this directory.
#
# The first field is the user's name and can be up to
# 253 characters in length. This is followed (on the same line) with
# the list of authentication requirements for that user. This can
# include password, comm server name, comm server port number, protocol
#
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify

```

- 2.7 For experiment, configure different views for TACACS+ server, RADIUS server and local login.

- 2.8 When done, enter 'exit' until you are disconnected.
- 2.9 Now, login again using console connection or SSH. What is your assigned view after login?
- 2.10 What are the commands available to you?
- 2.11 Simulate TACACS+ server is down by disconnecting it from the switch, then try login again. What is your assigned view after login?
- 2.12 Next, simulate RADIUS server is down also by disconnecting it from the switch, then try login again. What is your assigned view after login?